

Evaluating String Theory through the Lens of Scientific Realism

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Introduction

String theory describes elementary particles as one-dimensional vibrating **strings**, which have no inner structure.

Definition.

Strings can either form closed loops (**closed strings**) or have two distinct endpoints (**open strings**).



Figure: Closed String and Open String.

Settings

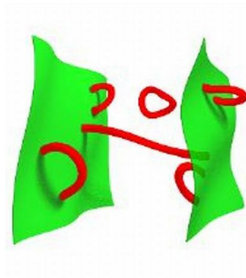


Figure: *D*-brane (created by Steuard Jensen).

Definition.

- A ***Dp-brane*** is a p -dimensional object embedded in a D -dimensional spacetime.
- D represents the endpoints of the open string satisfy Dirichlet boundary conditions:

$$X^I \Big|_{\sigma=0,\pi} = c^I,$$

where X^I represents the spacetime coordinates perpendicular to the brane, and σ is the spatial coordinate on the string from one endpoint ($\sigma = 0$) to the other endpoint ($\sigma = \pi$).

Examples.

- $p = 0$: **point particle** traces a **worldline**.
- $p = 1$: **string** traces a **worldsheet**.
- $p = 2$: **membrane** sweeps a **worldvolume**.

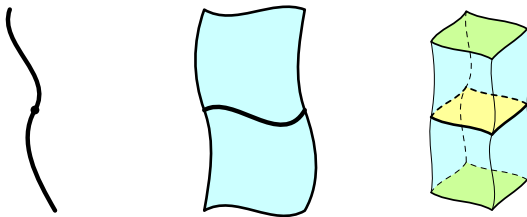


Figure: Worldline, Worldsheet, and Worldvolume.

Gauge Bosons

The **Polyakov action** for open strings is:

$$S = -\frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}, \quad g \equiv \det g,$$

which leads to a wave function. Applying the boundary conditions to the general solution leads to the mode expansion:

$$X^\mu(\tau, \sigma) = x^\mu + 2\alpha' p^\mu \tau + i\sqrt{2\alpha'} \sum_{n \neq 0} \frac{1}{n} \alpha_n^\mu e^{-in\tau} \cos(n\sigma).$$

Using the **Virasoro constraints**:

$$M^2 = \frac{1}{\alpha'} (N - 1),$$

where $N = \sum_{n>0} n \alpha_{-n}^\mu \alpha_n^\mu$. For $N = 1$, $M^2 = 0$ corresponds to a massless spin-1 gauge boson.

Gravitons

The **Polyakov action** for closed strings is:

$$S = -\frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}, \quad h = \det h_{\alpha\beta}$$

with periodic boundary conditions $X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma + 2\pi)$. The mode expansion is

$$X^\mu(\tau, \sigma) = X^\mu + 2\alpha' p^\mu \tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \left(\alpha_n^\mu e^{-in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-in(\tau+\sigma)} \right).$$

The Virasoro constraints lead to

$$M^2 = \frac{4}{\alpha'} (N - 1),$$

where $N = \sum_{n>0} \alpha_{-n}^\mu \alpha_n^\mu = \tilde{N}$. For $N = 1$, $M^2 = 0$ corresponds to a massless spin-2 particle, interpreted as the graviton.

Results

Conclusion.

- **Gauge bosons** arise from the massless states of open strings.
- **Gravitons** arise from the massless states of closed strings.

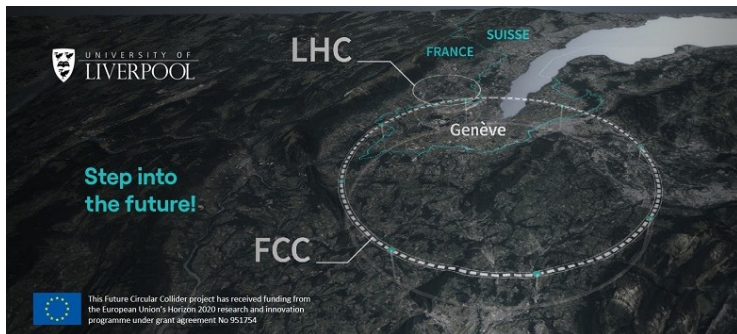
Strings that can vibrate in various modes, with each mode corresponding to a different particle type in spacetime, unifying **gauge interactions** (e.g., electromagnetism, weak, and strong forces) with **gravity**.

Remarks

- Potential *theory of everything*: unifies quantum mechanics and general relativity, encompass all fundamental forces.
- Incorporates several potential ideas beyond the standard model of physics (**supersymmetry** and **axions**).
- Offers new perspectives and tools in mathematics (**Calabi-Yau manifolds**).

However

- **Observability** typically extends to phenomena detectable with instruments.
- While we might place hope in the LHC or future projects like the FCC, string theory remain lacking experimental evidence.



Definition

Definition Scientific Realism

Science aims to give us a true picture of what the world looks like.

Problem

- Strings are unobservable
- Different string theories

Underdetermination

Different theories work equally well

Solving the problem (?)

- Dualities
- T-duality
- S-duality
- U-duality
- M-theory

Future Work

- **Ontology.** Entities like strings and branes are defined mathematically rather than empirically. The absence of a clear ontological foundation in string theory undermines its claims to describe reality.
- **Structural Realism.** Scientific theories tell us only about the mathematical structure of the world, not about the nature of objects.
- **Testable Predictions.** The predictive power of string theory may be reduced by the irreducibly statistical nature of quantum physics.
- ...

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